Wasiur R. KhudaBukhsh

Assistant Professor, School of Mathematical Sciences, University of Nottingham

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Appointments

Oct 2021 – present	**Assistant Professor** School of Mathematical Sciences, University of Nottingham, UK
Jan 2020 – Sep 2021	**President's Postdoctoral Scholar** Mathematical Biosciences Institute and College of Public Health, The Ohio State University, Columbus, OH, USA
Oct 2018 – Dec 2019	**Postdoctoral Researcher** Mathematical Biosciences Institute and College of Public Health, The Ohio State University, Columbus, OH, USA
Jan 2014 – Sep 2018	**Research Associate** Department of Electrical Engineering and Information Technology, Technische Universität Darmstadt, Germany
Jun 2011 – Dec 2013	**Manager, Advanced Analytics** Business Intelligence Unit, ICICI Bank, Mumbai, India

Education

2023 – 2024	**Postgraduate Certificate in Higher Education (PGCHE)** University of Nottingham, UK
2023	**Fellow of the Higher Education Academy (FHEA)** Advance HE, UK
2014 – 2018	**PhD (Dr. rer. nat.), summa cum laude** Department of Electrical Engineering and Information Technology, Technische Universität Darmstadt, Germany Thesis: *Model reductions for queueing and agent-based systems with applications in communication networks*. Advisor: Heinz Koeppl
2009 – 2011	**Master of Statistics (M.Stat.)** Indian Statistical Institute, Kolkata, India
2006 – 2009	**BSc (Honours) in Statistics** University of Calcutta, Kolkata, India

Research interests

Probability theory and stochastic processes; mathematical biology.

Grants and awards

Research funding

2024 – 2025 Engineering and Physical Sciences Research Council (EPSRC), EP/Y027795/1, principal investigator, £79,987.58 (March 2024 – February 2025)

Smaller grants and travel awards

- Isaac Newton Institute (INI) Retreat, 29 June – 3 July 2026 (with Olga Izyumtseva and Ercan Sönmez)
- Center for Mathematical Machine Learning and its Applications (CM2LA) Seorabol Research Station Workshop grant, South Korea, June 2025 (US$7,000)
- Isaac Newton Institute (INI) Retreat, 29 July – 4 August 2024
- London Mathematical Society (LMS) Research in Pairs grant, 2024 (£1,200)
- MATRIX–Simons Travel Grant, February 2024 (AU$2,500)
- University of Nottingham Faculty of Science International Research Collaboration Fund (£3,150)
- Heilbronn Institute for Mathematical Research (HIMR) Small Grants Scheme (£1,000)
- Royal Society International Exchange grant, 2022 (£11,702.09)
- University of Nottingham Faculty of Science International Collaboration Fund (£4,375)
- LMS Undergraduate Research Bursary, 2022 (£900, matched by £900 from the University of Nottingham)
- LMS Research in Pairs grant, 2022 (£1,200)
- Institute of Mathematical Statistics New Researchers Conference, 2022 (US$1,860)
- American Institute of Mathematics (AIM) SQuaREs (Structured Quartet Research Ensembles), with Hye-Won Kang, Lea Popovic, Grzegorz A. Rempała, Ruth J. Williams, and Felipe Campos
- German Academic Exchange Service (DAAD) travel grant to attend the Annual Meeting of the SMB, Sydney, Australia, 2018 (c. €2,000)

Awards

2024 Best Paper Award, International Conference on Quantum Communications, Networking, and Computing (QCNC 2024)

2019 President's Postdoctoral Scholars Program (PPSP), The Ohio State University

2018 Best-in-Session Presentation Award, IEEE International Conference on Computer Communications (INFOCOM), Honolulu, HI, USA

Publications

In mathematics, authors are conventionally listed alphabetically.

Preprints and submitted papers

1. Olga Izyumtseva and **Wasiur R. KhudaBukhsh**. Self-intersection local times of Volterra Gaussian bridges with large end-to-end distance. 2026. Submitted.
2. **Wasiur R. KhudaBukhsh** and Yangrui Xiang. Cutoff profiles and boundary effects in the togashi–kaneko model of autocatalytic reactions with weak mutations. 2026. Submitted.
3. Kushankur Dutta, Olga Izyumtseva, **Wasiur R. KhudaBukhsh**, and Grzegorz A. Rempała. Coloured epidemic models: functional law of large numbers and propagation of chaos. 2026. Submitted.

4. Olga Izyumtseva, **Wasiur R. KhudaBukhsh**, M. Gabriela M. Gomes, and Grzegorz A. Rempała. From individual-based stochastic epidemics to heterogeneous SIR equations. 2026. Minor revisions requested.
5. Yordan P. Raykov, Hengrui Luo, Justin D. Strait, and **Wasiur R. KhudaBukhsh**. Shrinkage priors for Bayesian substitute confounders. 2026.
6. Arnab Ganguly and **Wasiur R. KhudaBukhsh**. Fluctuation analysis of the standard QSSA for a stochastic multistage Michaelis–Menten model. 2026. Submitted.
7. Olga Izyumtseva and **Wasiur R. KhudaBukhsh**. Self-intersection local times for Volterra Gaussian processes in stochastic flows with interaction. 2026. Submitted.
8. J. D. Harborne, **Wasiur R. KhudaBukhsh**, and John R. King. Deterministic and stochastic asymptotics for a multiscale epidemic model. 2026. Revision requested.
9. Yordan P. Raykov, Hengrui Luo, Justin D. Strait, and **Wasiur R. KhudaBukhsh**. Kernel-based estimators for functional causal effects. 2026.
10. **Wasiur R. KhudaBukhsh** and Yangrui Xiang. Mixing time for a noisy SIS model on graphs. 2026. Submitted.
11. Karim S. Elsayed, Olga Izyumtseva, **Wasiur R. KhudaBukhsh**, and Amr Rizk. Stochastic analysis of entanglement-assisted quantum communication channels. 2025. Revision requested.
12. Olga Izyumtseva and **Wasiur R. KhudaBukhsh**. Local times of self-intersection and sample path properties of Volterra Gaussian processes. 2025. Minor revisions requested.

Peer-reviewed journal articles

13. Arnab Ganguly and **Wasiur R. KhudaBukhsh**. Statistical inference for a multiscale stochastic model of enzyme kinetics via propagation of chaos. *Stochastic Processes and their Applications*, 203:105122, 2027.
14. Arnab Ganguly and **Wasiur R. KhudaBukhsh**. Asymptotic analysis of the total quasi-steady state approximation for the Michaelis–Menten enzyme kinetic reactions. *Journal of Mathematical Analysis and Applications*, 561(1):130551, 2026.
15. Yi Fu, Hye-Won Kang, **Wasiur R. KhudaBukhsh**, Lea Popovic, Grzegorz A. Rempała, and Ruth J. Williams. Fragility in a Togashi–Kaneko stochastic model with mutations. *SIAM Journal on Life Sciences*, 1(2):202–228, 2026.
16. Riccardo Corradin, Luca Danese, **Wasiur R. KhudaBukhsh**, and Andrea Ongaro. Model-based clustering of time-dependent observations with common structural changes. *Statistics and Computing*, 36(7), 2026.
17. Alexander E. Zarebski, Nefel Tellioglu, Jessica E. Stockdale, Julie A. Spencer, **Wasiur R. KhudaBukhsh**, Joel C. Miller, and Cameron Zachreson. Including frameworks of public health ethics in computational modelling of infectious disease interventions. *Interface Focus*, 15, 2025.
18. Yushuf Sharker, Zaynab Diallo, **Wasiur R. KhudaBukhsh**, and Eben Kenah. Pairwise accelerated failure time regression models for infectious disease transmission in close contact groups with external sources of infection. *Statistics in Medicine*, 2024.
19. **Wasiur R. KhudaBukhsh** and Grzegorz A. Rempała. How to *correctly* fit an SIR model to data from an SEIR model? *Mathematical Biosciences*, 2024.
20. Matthew Wascher, Patrick Schnell, **Wasiur R. KhudaBukhsh**, Mikkel B.M. Quam, Joseph Tien, and Grzegorz Rempała. Estimating disease transmission in a closed population under repeated testing. *Journal of the Royal Statistical Society: Series C (JRSSC)*, 2024.
21. István Z. Kiss, Luc Berthouze, and **Wasiur R. KhudaBukhsh**. Towards inferring network properties from epidemic data. *Bulletin of Mathematical Biology*, 2024.

22. **Wasiur R. KhudaBukhsh**, Sat Kartar Khalsa, Eben Kenah, Grzegorz Rempała, and Joseph Tien. COVID-19 dynamics in an Ohio prison. *Frontiers in Public Health*, 2023.
23. **Wasiur R. KhudaBukhsh**, Caleb Deen Bastian, Matthew Wascher, Colin Klaus, Saumya Yashmohini Sahai, Mark H. Weir, Eben Kenah, Elisabeth Root, Joseph H. Tien, and Grzegorz A. Rempała. Projecting COVID-19 cases and hospital burden in Ohio. *Journal of Theoretical Biology*, 561:111404, 2023.
24. Colin Klaus, Matthew Wascher, **Wasiur R. KhudaBukhsh**, and Grzegorz Rempała. Likelihood-free dynamical survival analysis applied to the COVID-19 epidemic in Ohio. *Mathematical Biosciences and Engineering*, 20, 2023.
25. Kai Cui, **Wasiur R. KhudaBukhsh**, and Heinz Koeppl. Hypergraphon mean-field games. *Chaos*, 2022.
26. **Wasiur R. KhudaBukhsh**, Casper Woroszylo, Grzegorz Rempała, and Heinz Koeppl. A functional central limit theorem for SI processes on configuration model graphs. *Advances in Applied Probability*, 2022.
27. Colin Klaus, Matthew Wascher, **Wasiur R. KhudaBukhsh**, Joseph H. Tien, Grzegorz A. Rempała, and Eben Kenah. Assortative mixing among vaccination groups and biased estimation of reproduction numbers. *The Lancet Infectious Diseases*, 22:P579–581, 2022.
28. Francesco Di Lauro*, **Wasiur R. KhudaBukhsh***, István Z. Kiss, Eben Kenah, Max Jensen, and Grzegorz Rempała. Dynamic survival analysis for non-Markovian epidemic models. *Journal of the Royal Society Interface*, 2022. *Joint first authors.
29. Kai Cui, **Wasiur R. KhudaBukhsh**, and Heinz Koeppl. Motif-based mean-field approximation of interacting particles on clustered networks. *Physical Review E*, 105, 2022.
30. Harley Vossler, Pierre Akilimali, Yuhan Pan, **Wasiur R. KhudaBukhsh**, Eben Kenah, and Grzegorz A. Rempała. Analysis of individual-level epidemic data: study of 2018–2020 Ebola outbreak in Democratic Republic of the Congo. *Scientific Reports*, 12, 2022.
31. Ido Somekh*, **Wasiur R. KhudaBukhsh***, Elisabeth Dowling Root*, Greg Rempala, Eric Simoes, and Eli Somekh. Quantifying the population-level effect of COVID-19 mass vaccination campaign in Israel: a modeling study. *Open Forum Infectious Diseases*, 2022. *Joint first authors.
32. **Wasiur R. KhudaBukhsh**, Hye-Won Kang, Eben Kenah, and Grzegorz Rempała. Incorporating age and delay into models for biophysical systems. *Physical Biology*, 18(1), 2021. Invited paper.
33. **Wasiur R. KhudaBukhsh**, Boseung Choi, Eben Kenah, and Grzegorz Rempała. Survival dynamical systems: individual-level survival analysis from population-level epidemic models. *Journal of the Royal Society Interface Focus*, 10(1), 2020.
34. **Wasiur R. KhudaBukhsh**, Arnab Auddy, Yann Disser, and Heinz Koeppl. Approximate lumpability for Markovian agent-based models using local symmetries. *Journal of Applied Probability*, 56, 2019.
35. Hye-Won Kang*, **Wasiur R. KhudaBukhsh***, Heinz Koeppl, and Grzegorz Rempała. Quasi-steady-state approximations derived from a stochastic enzyme kinetics. *Bulletin of Mathematical Biology*, 81(5):1303–1336, 2019. *Joint first authors.
36. Saumya Yashmohini Sahai, Saket Gurukar, **Wasiur R. KhudaBukhsh**, Srinivasan Parthasarathy, and Grzegorz A. Rempała. A machine learning model for nowcasting epidemic incidence. *Mathematical Biosciences*, 2021.
37. **Wasiur R. KhudaBukhsh**, Sounak Kar, Bastian Alt, Amr Rizk, and Heinz Koeppl. Generalized cost-based job scheduling in very large heterogeneous cluster systems. *IEEE Transactions on Parallel and Distributed Systems*, 31(11):2594–2604, 2020.
38. Boseung Choi, Sydney Busch, Dieudonné Kazadi, Benoit Ilunga, Emile Okitolonda, Yi Dai, Robert Lumpkin, Omar Saucedo, **Wasiur R. KhudaBukhsh**, Joseph Tien, Marcel Yotebieng,

Eben Kenah, and Grzegorz A. Rempała. Modeling outbreak data: analysis of a 2012 Ebola virus disease epidemic in DRC. *BIOMATH*, 8(2), 2019.

39. **Wasiur R. KhudaBukhsh**, Amr Rizk, Sounak Kar, and Heinz Koeppl. Provisioning and performance evaluation of parallel systems with output synchronization. *ACM Transactions on Modeling and Performance Evaluation of Computing Systems (TOMPECS)*, 4(1), 2019.
40. Bastian Alt, Markus Weckesser, Christian Becker, Matthias Hollick, Sounak Kar, Anja Klein, Robin Klose, Roland Kluge, Heinz Koeppl, Boris Koldehofe, **Wasiur R. KhudaBukhsh**, Mahdi Mousavi, Martin Pfannemueller, Amr Rizk, Andy Schuerr, and Ralf Steinmetz. Transitions: a protocol-independent view of the future internet. *Proceedings of the IEEE*, 107(4):835–846, 2019.

Book chapters

41. Olga Izyumtseva, **Wasiur R. KhudaBukhsh**, and Grzegorz A. Rempała. Functional law of large numbers for an epidemic model with random effects. Handbook of Statistics. Elsevier, 2024.
42. Grzegorz A. Rempała and **Wasiur R. KhudaBukhsh**. Dynamical survival analysis for epidemic modeling. In *Handbook of Visual, Experimental and Computational Mathematics*, pages 1–17. Springer International Publishing, 2023.

Peer-reviewed conference proceedings

43. Karim Elsayed, **Wasiur R. KhudaBukhsh**, and Amr Rizk. On the trade-off between fidelity and latency for the quantum link layer with few memories and entanglement purification. In *Proceedings of the International Conference on Quantum Communications, Networking, and Computing (QCNC 2024)*, 2024. Best Paper Award.
44. Karim Elsayed, **Wasiur R. KhudaBukhsh**, and Amr Rizk. On the fidelity distribution of link-level entanglements under purification. In *Proceedings of the IEEE International Conference on Communication (ICC) 2024*, 2024.
45. Riccardo Corradin, Luca Danese, **Wasiur R. KhudaBukhsh**, and Andrea Ongaro. Model-based clustering of non-stationary time series with common historical change times. In *Statistical Learning, Sustainability and Impact*, 2023.
46. **Wasiur R. KhudaBukhsh**, Bastian Alt, Sounak Kar, Amr Rizk, and Heinz Koeppl. Collaborative uploading in heterogeneous networks: optimal and adaptive strategies. In *IEEE International Conference on Computer Communications (INFOCOM)*, 2018. $<$20% acceptance rate. Best-in-Session Presentation Award.
47. **Wasiur R. KhudaBukhsh**, Amr Rizk, Alexander Frömmgen, and Heinz Koeppl. Optimizing stochastic scheduling in fork-join queueing models: bounds and applications. In *IEEE International Conference on Computer Communications (INFOCOM)*, 2017. $\sim$20% acceptance rate.
48. Adrian Šošić, **Wasiur R. KhudaBukhsh**, A. M. Zourbir, and Heinz Koeppl. Inverse reinforcement learning in swarm systems. In *AAMAS Workshop on Transfer in Reinforcement Learning*, 2017. Available: http://www.tirl.info/proceedings/2017/SosicEtal-Inverse.pdf.
49. Adrian Šošić, **Wasiur R. KhudaBukhsh**, A. M. Zourbir, and Heinz Koeppl. Inverse reinforcement learning in swarm systems. In *International Conference on Autonomous Agents & Multiagent Systems (AAMAS)*, 2017. $\sim$26% acceptance rate. Best Paper Award finalist.
50. **Wasiur R. KhudaBukhsh**, Julius Rückert, Julian Wulfheide, David Hausheer, and Heinz Koeppl. Analysing and leveraging client heterogeneity in swarming-based live streaming. In *IFIP Networking Conference (IFIP Networking) and Workshops*, 2016. $\sim$26% acceptance rate.
51. Mahdi Mousavi, Hussein Al Shatri, **Wasiur R. KhudaBukhsh**, Heinz Koeppl, and Anja Klein. Cross-layer QoE-based incentive mechanism for video streaming in multi-hop wireless networks. In *IEEE 86th Vehicular Technology Conference (VTC)*, 2017.

Thesis and technical notes

52. **Wasiur R. KhudaBukhsh**. *Model reductions for queueing and agent-based systems with applications in communication networks*. PhD thesis, Technische Universität, Darmstadt, 2018. Available at: http://tuprints.ulb.tu-darmstadt.de/7588/.
53. Mark Sinzger-D'Angelo, Heinz Koeppl, and **Wasiur R. KhudaBukhsh**. Bounds on the spectral radius of real-valued non-negative kernels on measurable spaces. 2023. Preprint: https://arxiv.org/abs/1808.00258.

Talks

60 invited and 30 contributed or local talks; 7 poster presentations.

Invited talks

Jun 2026	Stochastic Processes and their Applications (SPA 2026), Ithaca, NY, USA
May 2026	Workshop "Individual variation and its consequences for population dynamics", University of Strathclyde, Glasgow, UK
Apr 2026	Probability Seminar, Louisiana State University, USA
Mar 2026	Mathematics in Life Sciences Seminar, University of Strathclyde, UK
Jan 2026	Department Seminar, School of Mathematical and Statistical Sciences, IIT Mandi, India
Dec 2025	International Conference on Recent Advances in Medical Science and Technology, IIT Kharagpur, India
Nov 2025	HEALMOD Seminar, The Ohio State University, USA (online)
Oct 2025	Statistics and Probability Seminar, University of Manchester, UK
Sep 2025	Workshop on Gaussian and Lévy-type stochastic processes, University of Nottingham, UK
Sep 2025	Annual Meeting of the Royal Statistical Society, Edinburgh, UK
Jul 2025	Stochastic Processes and their Applications (SPA 2025), Wrocław, Poland
Jun 2025	MIDAS Webinar (online)
Jun 2025	BIRS-CMO Workshop "Mathematical and statistical challenges in post-pandemic epidemiology and public health", Oaxaca, Mexico
May 2025	Statistics and Probability Seminar, University of Essex, UK
Dec 2024	Innovations in Machine Learning, Artificial Intelligence, Data Science and Modelling (IMLAIDSM'24), India
Nov 2024	Probability Seminar, Louisiana State University, USA
Nov 2024	Mathematics of Reaction Networks (MoRN) Seminar
Oct 2024	CMMB Seminar, University of Nottingham, UK
Jun 2024	Mini-symposium, Annual Meeting of the Society for Mathematical Biology, Seoul, South Korea
Jun 2024	Epidemiology Workshop, Konkuk University, Seoul, South Korea
Dec 2023	Mathematical Biology Seminar, University of Leeds, UK
Dec 2023	Mathematical Life Sciences talk, University of Bonn, Germany
Nov 2023	Probability Seminar, Louisiana State University, USA
Aug 2023	Mini-symposium, International Congress on Industrial and Applied Mathematics (ICIAM), Tokyo, Japan

Jul 2023	SMB Diversity in Math Bio seminar series
Jun 2023	Mini-symposium, SIAM Conference on Optimization, Seattle, WA, USA
Apr 2023	BBSRC DTP event, University of Nottingham, UK
Apr 2023	Probability Seminar, Louisiana State University, USA
Mar 2023	Stochastics Seminar, Georgia Tech, USA (online)
Feb 2023	NANT Lecture, Networking in Applied Network Theory, Będlewo, Poland (online)
Dec 2022	Theoretical Statistics and Mathematics Seminar, Indian Statistical Institute, New Delhi, India
Dec 2022	Workshop on Non-equilibrium Phenomena in Physics and Biology, Asia Pacific Center for Theoretical Physics, South Korea
Dec 2022	Self-Organizing Systems Workshop, Technische Universität Darmstadt, Germany
Apr 2022	Mini-symposium, SIAM Conference on Uncertainty Quantification
Feb 2022	Statistics and Probability Seminar, University of Nottingham, UK
Jan 2022	Mathematics of Reaction Networks (MoRN) Seminar
Nov 2021	Math Bio Seminar, The Ohio State University, USA (online)
Oct 2021	CMMB Seminar, University of Nottingham, UK
Oct 2021	Stochastic Systems Seminar, Department of Mathematics, University of California San Diego, USA (online)
Sep 2021	Guest lecture on infectious disease epidemiology, The Ohio State University, USA (online)
Jul 2021	American Institute of Mathematics (AIM) workshop “Limits and control of stochastic reaction networks”
Apr 2021	Colloquium, Department of Mathematics, Simon Fraser University, Canada (online)
Jan 2021	Special session, Joint Mathematics Meetings (JMM) (online)
Dec 2020	IDI Virtual COVID-19 Symposium, The Ohio State University, USA (online)
Nov 2020	Seminar, Institute of Applied Mathematics and Mechanics, University of Warsaw, Poland
Nov 2020	Seminar, Institute of Applied Mathematics and Mechanics, University of Warsaw, Poland
Oct 2020	Biostatistics Seminar, The Ohio State University, Columbus, OH, USA
Sep 2020	Math Bio Seminar, Virginia Tech, USA (online)
Sep 2020	International Webinar Series on Artificial Intelligence and Machine Learning (online)
Aug 2020	Workshop “Life on Planet Earth: Above and Below”, Mathematical Biosciences Institute, The Ohio State University, USA
Jul 2020	Special session on data science, Business Intelligence Unit, ICICI Bank, Mumbai, India
Jul 2020	International Webinar, Departments of Mathematics and Physics, Bidhan Chandra College, India (online)
May 2020	Special session, The Erdős Institute Python Bootcamp (with Marissa Renardy)
Feb 2020	Biostatistics Seminar, The Ohio State University, Columbus, OH, USA
Feb 2020	Mathematical Modeling and Statistical Analysis of Infectious Disease Outbreaks, CIRM, Marseille, France

2019	Mini-symposium, Annual Meeting of the Society for Mathematical Biology, Montreal, Canada
Jun 2019	Mini-symposium, International Conference on Mathematical Methods and Models in Biosciences (BioMath), Będlewo, Poland
May 2019	BIRS-CMO Workshop "Scaling Limits of Dynamical Processes on Random Graphs", Oaxaca, Mexico
2018	Mini-symposium, Joint Annual Meeting of the SMB and JSMB, Sydney, Australia
May 2017	Cloud Computing, Machine Learning and Networking Research (CLAN) Lab, Purdue University, USA (host: Vaneet Aggarwal)

Contributed talks

Dec 2026	MIDAS Webinar (online)
Jun 2026	Stochastic Processes and their Applications (SPA 2026), Ithaca, NY, USA
Sep 2025	Annual Meeting of the Royal Statistical Society, Edinburgh, UK
Jun 2024	Melbourne Mathematical Biology Seminar, University of Melbourne, Australia
Jun 2024	IEEE International Conference on Communications (ICC), Denver, CO, USA
Jul 2023	Mini-symposium, Annual Meeting of the Society for Mathematical Biology, Columbus, OH, USA
Jul 2023	Dynamical Systems in the Life Sciences, The Ohio State University, USA
Jun 2022	BIRS Workshop "Preparing for the next pandemic"
Jun 2021	Annual Meeting of the Society for Mathematical Biology
Jun 2021	Workshop on Mathematical and Computational Biology
May 2021	SIAM Conference on Applications of Dynamical Systems (DS21)
Mar 2021	ENAR Spring Meeting
Mar 2021	Mathematical Biosciences Institute (MBI) Seminar, Columbus, OH, USA
Feb 2021	Joint Mathematical Epidemiology and Math Education SMB Subgroup Meeting
Nov 2020	Applied Mathematics Seminar, Department of Mathematics, The Ohio State University, USA
Aug 2020	Bernoulli–IMS One World Symposium (online)
Aug 2020	Mini-symposium, Annual Meeting of the Society for Mathematical Biology (online)
Aug 2020	Joint Statistical Meetings (JSM) (online)
Jun 2020	Mini-symposium, SIAM Conference on Mathematics of Data Science (online)
May 2020	MBI Workshop on Mathematical and Computational Methods in Biology
Dec 2019	International Indian Statistical Association Conference (IISA 2019), Mumbai, India
Dec 2019	Latin American Congress of Probability and Mathematical Statistics (CLAPEM XV), Mérida, Yucatán, Mexico
Oct 2019	Mathematical Biosciences Institute (MBI) Seminar, Columbus, OH, USA
2019	Joint Statistical Meetings (JSM), American Statistical Association, Denver, CO, USA
Jul 2019	Stochastic Processes and their Applications, Bernoulli Society, Northwestern University, USA

Jun 2019	International Conference on Mathematical Methods and Models in Biosciences (BioMath), Poland
2019	Mathematical Biosciences Institute (MBI) Seminar, Columbus, OH, USA
2018	IEEE International Conference on Computer Communications (INFOCOM), Honolulu, HI, USA
2017	IEEE International Conference on Computer Communications (INFOCOM), Atlanta, GA, USA
2016	IFIP Networking Conference and Workshops, Vienna, Austria

Posters

Sep 2022	Annual Meeting of the Society for Mathematical Biology, Heidelberg, Germany
Aug 2022	22nd Meeting of New Researchers in Statistics and Probability, George Mason University, USA
Feb 2020	Mathematical Models in Evolutionary Biology, CIRM, Marseille, France
Sep 2019	Annual Meeting of the Infectious Diseases Institute, The Ohio State University, Columbus, OH, USA
2019	Joint Statistical Meetings (JSM), American Statistical Association, Denver, CO, USA
Jun 2019	CMCF Algorithms and Methods for Single Cell Genomics, University of California, Irvine, USA
May 2019	BIRS-CMO Workshop "Scaling Limits of Dynamical Processes on Random Graphs", Oaxaca, Mexico

Teaching

University of Nottingham, UK (module leader or co-lecturer)

2025 – 26	MATH3060 Financial Mathematics stream (with Michael Tretyakov)
2025, 2026	MATH2109 Probability 3 / Markov Chains (c. 250 and 270 students)
2022 – 2024	MATH2010 Probability Models and Methods (c. 270, 260 and 282 students)
2022 – 25	MATH4045 Mathematical Finance stream (2022–23 with Fabrizio Leisen; 2023–24 with Fabrizio Leisen and John Billingham; 2024–25 with Marco Iglesias)

The Ohio State University, USA (instructor)

Autumn 2019	STAT 4201 Introduction to Mathematical Statistics, Department of Statistics (over 120 students)
Summer 2019	PUBHBIO 7193 Individual Studies in Biostatistics (advanced stochastic epidemiology), Division of Biostatistics (with Eben Kenah)

Technische Universität Darmstadt, Germany (teaching assistant)

2016, 2017	Computational Methods for Systems and Synthetic Biology, Department of Electrical Engineering and Information Technology

Supervision

Supervised or co-supervised 17 MSc and 16 BSc students (not including REU students).

Postdoctoral researchers (University of Nottingham)

2023 – 2026 Olga Izyumtseva (British Academy Fellowship)

2024 – 2025 Yangrui Xiang

PhD students (University of Nottingham)

2023 – 2027 Kushankur Dutta (expected April 2027; secondary supervisors: Mikhail Tretyakov and Karthik Bharath)

2023 – 2026 Wesam El Buaishi (expected December 2026; primary supervisor: Yordan Raykov)

2022 – 2026 James Harborne (viva November 2026; primary supervisor: John King)

MSc students

2026 Jennifer Wood, MSc dissertation, University of Nottingham

2025 Wenhao Qiu, MSc dissertation, University of Nottingham

2025 Maxwell Hancock, MSc dissertation, University of Nottingham

2025 William Bateman, MSc dissertation, University of Nottingham

2024 Siyu Xu, MSc dissertation, University of Nottingham

2024 Qi Zhang, MSc dissertation, University of Nottingham

2024 Kaustav Chakraborty, MSc dissertation, University of Nottingham

2023 Nazreen Asok, MSc dissertation, University of Nottingham

2023 Changqing Du, MSc dissertation, University of Nottingham

2022 Yajie Guo, MSc dissertation, University of Nottingham

2022 Jianing Ye, MSc dissertation, University of Nottingham

2019 Robert Lumpkin, master's project, The Ohio State University (with Grzegorz A. Rempała; resulted in [38])

2018 Ranjani Krishnan, master's thesis, Technische Universität Darmstadt (with Denny Stohr)

2018 Sayantan Kumar, summer internship, Technische Universität Darmstadt

2017 Arnab Auddy, summer internship, Technische Universität Darmstadt (resulted in [34])

2017 Markus Schanz, master's thesis, Technische Universität Darmstadt (with Christian Koch)

2017 Hameer Abbasi, master's thesis, Technische Universität Darmstadt (with Bastian Alt)

BSc students

2025 Yiwen Yan, summer internship, University of Nottingham

2023 Maoduan Ran, summer internship, University of Nottingham

2022 Sarina Rivlin-Sanders, summer internship, University of Nottingham

2022 Junzhe Zhu, summer internship, University of Nottingham

2019 Jasen Lai, Ran An, Stanley L. Cao, Hongyi Wang, Joey Zeng, and Kai Li, summer projects, The Ohio State University

2019	Yi Dai and Jordan Kirsch, summer projects, The Ohio State University (with Grzegorz A. Rempała and Eben Kenah)
2018	Siddhaarth Sarkar, summer internship, Technische Universität Darmstadt
2017	Simon Schwanz, project seminar, Technische Universität Darmstadt (with Christian Koch)
2016	Vikash Vikash, summer internship, Technische Universität Darmstadt
2016	Simon Schwanz, proseminar, Technische Universität Darmstadt

Service

Editorial

2026 – present	Editorial Board Member, *Mathematical Biosciences*
2023	Guest Associate Editor (on behalf of the Applied Probability Section of the RSS), discussion paper for the *Journal of the Royal Statistical Society: Series B*
	Guest Editor, special issue of *Mathematical Biosciences and Engineering*

Committees

2025 – present	Secretary and Treasurer, Reaction Networks Subgroup of the Society for Mathematical Biology (SMB)
2023 – present	Committee member, Applied Probability Section of the Royal Statistical Society (RSS)

PhD examining

2025	Internal examiner: Liam Critcher, School of Mathematical Sciences, University of Nottingham, UK
2025	External examiner: Dániel Keliger, Institute of Mathematics, Budapest University of Technology and Economics, Hungary
2024	External examiner: Hiu Ching Yip, Politecnico di Torino, Italy

Grant review

2024	Engineering and Physical Sciences Research Council (EPSRC), UK Research and Innovation
2024	Netherlands Organisation for Health Research and Development (ZonMw)

Organising conferences and workshops

Jul 2026	Stochastic Reaction Networks subgroup mini-symposium, joint Annual Meeting of the ESMTB and SMB, Graz, Austria (with Stefan Müller)
Jul 2026	Mini-symposium, joint Annual Meeting of the ESMTB and SMB, Graz, Austria (with Olga Izyumtseva and Arnab Ganguly)
Jun 2026	Contributed session, Stochastic Processes and their Applications 2026, Cornell University, USA (with Olga Izyumtseva)
May 2026	Workshop “Recent advances in stochastic processes: theory and applications”, University of Edinburgh, UK (with Tess Vernon)
Sep 2025	Workshop “Gaussian and Lévy-type stochastic processes”, University of Nottingham, UK (with Olga Izyumtseva)

Jun 2025	BIRS Workshop “Mathematical and statistical challenges in post-pandemic epidemiology and public health” (with Jessica Stockdale, Sara Del Valle, Joel C. Miller, and Grzegorz A. Rempała)
Jul 2024	Satellite Workshop on Chemical Reaction Networks, Gyeongju, South Korea (with Hye-Won Kang and Jinsu Kim)
Jun 2024	Mini-symposium, Annual Meeting of the Society for Mathematical Biology, Seoul, South Korea (with Hye-Won Kang and Jinsu Kim)
Oct 2023	Workshop on Epidemic Modelling, Applied Probability Section of the Royal Statistical Society (with Alex Cox, Fraser Daly, and Lucy Teece)
Jul 2023	Mini-symposium, Annual Meeting of the SMB, Columbus, OH, USA (with Hye-Won Kang)
May 2023	Mini-symposium, SIAM Conference on Optimization, Seattle, WA, USA
Sep 2022	Workshop “Recent challenges in mathematical epidemiology”, University of Nottingham, UK (with Kirsty Bolton and Yordan Raykov)
Sep 2022	Mini-symposium, European Conference on Mathematical and Theoretical Biology (ECMTB 2022) (with Hye-Won Kang)
Jun 2022	BIRS Workshop “Preparing for the next pandemic” (with Sara Del Valle, Joel C. Miller, and Rick Durrett)
2021	Mini-symposium, SIAM Conference on Applications of Dynamical Systems
2021	Mini-symposium, Annual Meeting of the Society for Mathematical Biology (with Hye-Won Kang)
2021	Invited session, ENAR Annual Meeting (with Marissa Renardy)
2020	Mini-symposium, Annual Meeting of the Society for Mathematical Biology (with Hye-Won Kang)
Jun 2020	Mini-symposium, SIAM Conference on Mathematics of Data Science (MDS20), Cincinnati, OH, USA (with Veronica Ciocanel)
Dec 2019	Contributed session, Latin American Congress of Probability and Mathematical Statistics (CLAPEM XV), Mérida, Yucatán, Mexico (with Hye-Won Kang)
Dec 2019	Session, International Indian Statistical Association Conference (IISA 2019), Mumbai, India (with Arindam Fadikar)

Department and university

2022 – present	Course Director, BSc/MMath with a year in industry, University of Nottingham
2022 – 2023	Organiser, Statistics and Probability Seminars, University of Nottingham
2021	Reviewer, President’s Research Excellence awards, The Ohio State University
2019	Judge, OPA travel awards, The Ohio State University
2018, 2019	Judge, OSU Mathematical Contest in Modeling, The Ohio State University

COVID-19 modelling response

Member of the OSU/IDI COVID-19 response modelling team, providing daily predictions of COVID-19 cases and the resulting hospital burden in Ohio to the Ohio Department of Health and the Ohio Hospital Association.

Tutorials and REU mentoring

2020	"Modelling the COVID-19 pandemic", Erdős Institute bootcamp
2019	"How to write scientific papers", MBI REU students
2019	"Introduction to epidemic modelling", SAMMS REU students
2019	Mentor, MBI REU programme; Ohio 5 SURE students, College of Public Health, The Ohio State University

Refereeing

Electronic Journal of Probability
Journal of Applied Probability
Advances in Applied Probability
Stochastic Processes and their Applications
Stochastics
Statistics and Probability Letters
Methodology and Computing in Applied Probability
SIAM Journal on Applied Dynamical Systems
SIAM Multiscale Modeling & Simulation
Biometrika
Mathematical Biosciences
Bulletin of Mathematical Biology
Journal of Mathematical Biology
Journal of Theoretical Biology
Theoretical Population Biology
Mathematical Biosciences and Engineering
PLOS One
PLOS Computational Biology
Journal of Chemical Physics
Journal of Mathematical Physics
Journal of Mathematical Analysis and Applications
Journal of Mathematics in Industry
Journal of Scientific Research
Afrika Matematika
Mathematics (MDPI)
European Journal of Control
IEEE Transactions on Automatic Control
IEEE Transactions on Networking
IEEE Transactions on Control of Network Systems
IEEE Control Systems Letters
IEEE Multi-Conference on Systems and Control (2015)
American Control Conference (ACC)

Reviewing and judging

	Reviewer for AMS Mathematical Reviews (MathSciNet) and zbMATH Open
2021, 2023	Judge, poster sessions, Annual Meeting of the Society for Mathematical Biology
2021	Judge, abstract submissions, Annual Meeting of the MIDAS Network

Memberships

Royal Statistical Society (Applied Probability Section committee member)
Society for Mathematical Biology (life member)

Media

- The Dynamical Systems Web (DSWeb): student feature, January 2021
- OSC Research Report: "COVID-19 projection models equip Ohioans"
- OSC News and HPCwire: "Pandemic modeling: COVID-19 projection models equip Ohioans"
- IDI News: "Far from trivial: a postdoctoral researcher's path to a key position in the OSU/IDI COVID-19 modeling team"
- STEAM Factory: member spotlight

Languages and computing

Languages	English (professional proficiency), Bengali (native), Hindi (working knowledge), German (intermediate, B1)
Computing	C, R, Python, SQL, Julia, MATLAB, SAS, LaTeX

References

Available on request.